Experiment Matrix

1. Purpose

This matrix organizes future experiments by research question, input type, and success metric.

The current comparison snapshot used in the project ranks the screening models as:

1. multimodal_attention
1. transformer
1. vit
1. cnn
1. lstm

2. Multimodal Screening Experiments

Experiment	Inputs	Model	Metric	Why It M
CNN baseline	handwriting + audio + errors	`InitialCNNModel`	accuracy / F1	Smallest
Text baseline	text + behavior + errors	`InitialLSTMModel`	accuracy / F1	Checks te
Full baseline	image + audio + text + behavior + errors	`InitialCNNLSTMModel`	accuracy / F1	Early full
Default multimodal	all modalities	`MultimodalDyslexiaModel`	accuracy / F1	Main pro
Transformer fusion	all modalities	`TransformerMultimodalModel`	accuracy / F1	Tests stro
Attention fusion	all modalities	`AttentionMultimodalModel`	accuracy / F1 + attention analysis	Tests inte
ViT fusion	all modalities	`ViTMultimodalModel`	accuracy / F1	Tests pat

3. Low-Resource Transfer Experiments

Experiment	Inputs	Method	Metric	Why It Matters
Scratch vs transfer	Bengali manifest	cross-lingual warm start	delta F1	Measures transfer benefit
Freeze vs finetune	Bengali manifest + source checkpoint	prefix freezing	delta F1	Finds stable warm-start recipe
Distillation test	Bengali manifest + teacher checkpoint	shared feature distillation	delta F1 / stability	Tests teacher-guided adaptati
Foundation adaptation	Bengali manifest	foundation + adapter	accuracy / F1	Tests reusable latent space

4. Explainability Experiments

Experiment	Inputs	Method	Metric	Why It Matters
Grad-CAM check	handwriting images	`GradCAM`	human plausibility review	Tests image explanation quality
Attention check	multimodal outputs	modality attention	agreement score	Tests if weights are sensible
Text attention check	text sequences	transformer token scores	qualitative review	Tests token-level interpretability
Educational summary review	prediction output	explanation text	teacher usability feedback	Tests whether explanations are reac

5. Biomarker Experiments

Experiment	Inputs	Method	Metric	Why It Matters
Biomarker ranking	numeric manifest	`discover_digital_biomarkers`	ranking stability	Finds strongest signals
Family comparison	feature families	group analysis	effect size	Compares handwriting vs speech vs reading
Cross-language comparison	Bengali + English	split-wise analysis	rank overlap	Tests portability of markers

6. Intervention Experiments

Experiment	Inputs	Method	Metric	Why It Matters
Plan selection	learner profile	`InterventionPolicy`	chosen action consistency	Tests policy behavior
Reward update	session progress	policy update loop	reward improvement	Tests learning signal
Tutor adaptation	tutor state	`AdaptiveTutorAgent`	session score gain	Tests adaptive practice value

7. Suggested Reporting Metrics

- accuracy
- F1 score
- recall for elevated-risk cases
- MAE for regression tasks
- explanation usefulness
- modality-attention stability
- biomarker rank stability
- intervention improvement across sessions
- cross-validation selection accuracy
- hard holdout F1 / precision / recall